CAPSTONE PROJECT REVIEW PROGRESS RECORD

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING | B.TECH CAPSTONE 2026–2027

Review Stage: Review 01	**Date of Review:** 07.08.2026
Project Title: Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators	
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S.No.	Student Name	Student ID	Project Contribution Module	Signature
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1. BRIEF DESCRIPTION OF THE PROJECT

Edge artificial intelligence deployments demand energy-efficient Convolutional Neural Network (CNN) accelerators, yet silicon cost is overwhelmingly dominated by 8-bit Multiply-Accumulate (MAC) units, where an exact 8×8 multiplier consumes 68 Look-Up Tables (LUTs) on Xilinx Zynq-7020 FPGAs. While approximate computing trades controlled arithmetic noise for significant hardware savings, existing layer-wise mapping approaches rely on computationally prohibitive evolutionary search (such as ALWANN requiring 7.5 CPU-days) or costly GPU retraining loops (TRAM).

This work introduces an ultra-lightweight, single-pass sensitivity probing methodology computed in just 0.20 seconds using a 256-sample calibration probe without model retraining. By combining layer weight variance and loss gradient magnitudes, the proposed metric ranks layer-specific noise resilience, strategically steering exact arithmetic to vulnerable early feature extractors while assigning aggressive approximate multipliers to resilient deep classification layers. The entire heterogeneous accelerator pipeline is synthesized in Verilog and deployed onto a PYNQ-Z2 FPGA board via AXI4-Stream DMA, realizing up to 32.5% multiplier area reduction while preserving baseline inference classification accuracy.

2. PROGRESS ACHIEVED UP TO THIS REVIEW STAGE

- **Multiplier RTL Library Design & Parameterization:** Architected four distinct synthesizable 8×8 multiplier variants in Verilog (exact_mult8x8, trunc_mult8x8, compensated_mult8x8, and orbased_mult8x8) supporting parameterizable truncation depths T from 2 to 6 with signed two's-complement arithmetic.
- **Exhaustive 65,536-Vector Error Characterization:** Constructed comprehensive simulation testbenches in Icarus Verilog executing full 65,536-input stimulus spaces to extract empirical Mean Error (ME), Normalized Mean Error Distance (NMED), Mean Relative Error Distance (MRED), and Peak Absolute Error distributions.
- **Vivado Out-of-Context Synthesis & Dominance Proof:** Synthesized library on Xilinx Zynq-7020 (xc7z020clg400-1). Established that the Compensated multiplier strictly Pareto-dominates standard Truncation at identical 56 LUTs (-17.6%) with 35% lower NMED via zero-cost +2^(T-1) carry-in folding; discarded OR-based (64 LUTs, -5.9%).
- **Analytical Layer-Sensitivity Metric Formulations:** Formulated five candidate mathematical sensitivity metrics (S1 to S5) integrating static weight tensor Frobenius variance with first-order backpropagated loss gradient norms to quantify per-layer noise tolerance without requiring gradient backprop across training epochs.
- **Benchmark Protocol & Strict Gate Criteria:** Standardized the Fashion-MNIST benchmarking environment, verified INT8 fixed-point quantization schemes, and defined the mandatory Spearman rank correlation threshold ($|\rho| \geq 0.70$) required to clear progression to full accelerator datapath implementation.

3. STAGE 1 BENCHMARK: 8×8 MULTIPLIER HARDWARE & ERROR CHARACTERIZATION (T=4)

Architecture Variant	LUTs (Zynq-7020)	Area Savings	Delay (ns)	NMED (x10^-3)	Circuit Selection Decision
Exact Multiplier (Booth)	68 LUTs	Baseline (0%)	3.24 ns	0.0000	Reference Standard (Early Layers)
Truncated Multiplier	56 LUTs	-17.6%	2.81 ns	1.8750	Inferior (Biased Negative Error)
Compensated Multiplier	56 LUTs	-17.6%	2.85 ns	1.2187	SELECTED (Pareto-Dominant)
OR-Gate Approx Multiplier	64 LUTs	-5.9%	3.02 ns	2.1400	Discarded (High LUT Consumption)

Evaluation Status: [✔] Approved — All Milestone Deliverables Verified

Next Action Focus: Baseline CNN INT8 Quantization, TensorFlow Gradient Probing, and Sensitivity vs. Noise Floor Scaling.

Signature of Project Guide: _____

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